MSD MANUAL
Veterinary Manual

PROFESSIONAL VERSION

Behavior Problems of Dogs

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Diagnosis of Behavior Problems of Dogs

- History
- Clinical examination

The process of diagnosing behavior problems is described in Diagnosis of Behavior Problems in Animals.

When assessing behavior problems in dogs, veterinarians must distinguish normal but undesirable behaviors from abnormal behaviors that are due to an underlying behavior disorder. Many behaviors, such as pulling, digging, destructive chewing, urine marking, predation, and play biting, are part of the normal canine behavior repertoire but are, nevertheless, highly undesirable. Problem behaviors that arise from inadvertent reinforcement, such as attention-seeking barking, food begging, and object stealing, are also normal, though undesirable.

When normal dog behaviors occur in excess, they can be disruptive enough to present serious challenges for their owners. Behavior problems rooted in fear, anxiety, excessive arousal, and impulsivity—such as fear-related aggression, possessive aggression, repetitive movement disorders, separation anxiety, and noise phobias—can develop as a result of genetic factors, stressful perinatal (prenatal, neonatal) environment, insufficient early socialization, medical conditions affecting brain health and development, and/or traumatic environmental events.

It is essential to rule out possible medical causes or factors contributing to the behavior problem. Pain, in particular, is an important risk factor for behavior problems. A study of veterinary behaviorists' cases found that between 28% and 82% of patients demonstrated signs of pain (1).

If medical causes have been excluded on the basis of thorough medical evaluation, diagnosis of behavior problems is based on a careful history, including observation of the dog's behavior and an assessment of the dog's physical and behavioral health.

Treatment of Behavior Problems of Dogs

- Environmental management
- Behavior modification
- Behavioral pharmacotherapy

Before a specific treatment to manage, improve, or resolve a behavior problem in a dog is implemented, some common elements that apply to most cases should be considered.

The initial discussion should focus on the following:

- reviewing the cause, diagnosis, and motivation behind the behavior
- providing the owner with a prognosis and realistic expectations for both short- and long-term outcomes
- ensuring that all of the dog's needs are being adequately met
- providing the owner with an understanding of normal behavior as it relates to the problem
- teaching the owner to read canine body language and facial expressions
- reviewing the principles of learning and reinforcement-based training (predictable consequences)
- managing both the environment and the dog to prevent further incidents

For common pharmacotherapeutic treatments, see the table Common Drugs Used to Treat Behavior Problems in Dogs.

TABLE

Common Drugs Used to Treat Behavior Problems in Dogs

Drug (Class)	Dosage	Indications	Adverse Effects
Fluoxetine (SSRI)	1–2 mg/kg, PO, q 24 h[a,b,c,d]	FDA approved for canine separation anxiety. Extralabel indications include anxiety disorders, phobias, conflict aggression, fear aggression, compulsive disorders, excessive arousal, impulsivity, and urine marking.[b,d,e]	Anorexia, hyporexia, lethargy, sedation, vomiting, diarrhea, restlessness, vocalization, agitation, tremors, seizures, changes in urine frequency. Aggression has also been reported. In humans, hyponatremia, changes in glucose, bleeding, and hepatotoxicosis have been reported.
Paroxetine (SSRI)	1–2 mg/kg, PO, q 24 h[b,d,e]	Anxiety disorders, phobias, conflict aggression, fear aggression, compulsive disorders, excessive arousal, impulsivity, and urine marking.[b,d,e] Extralabel.	Lethargy, salivation, hyporexia, anorexia, vomiting, diarrhea, anxiety, irritability, aggression, insomnia, hyponatremia, mydriasis, hyperactivity or panting, constipation, urinary retention, cardiac arrhythmias. Less likely than fluoxetine to cause anorexia. In humans, abnormal bleeding and hepatotoxicosis have been reported.
Sertraline (SSRI)	0.5–4 mg/kg, PO, q 24 h[b,c,d,e]	Anxiety disorders, phobias, conflict aggression, fear aggression, compulsive disorders, excessive arousal, impulsivity, and urine marking.[b,d,e] Extralabel.	Anorexia, hyporexia, sedation, lethargy, vomiting, diarrhea, changes in urine frequency, excitement, anxiety, agitation, seizures. In humans, abnormal bleeding and hepatotoxicosis have been reported.
Clomipramine (TCA)	Label dose: 2–4 mg/kg, PO, q 24 h, or 1–2 mg/kg, PO, q 12 h Extralabel dose: 1–4 mg/kg, PO, q 12–24 h[e,f]	FDA approved for canine separation anxiety. Extralabel indications include anxiety disorders, phobias, conflict aggression, fear aggression, compulsive disorders, excessive arousal, impulsivity, and urine marking.[b,d,e]	Anorexia, hyporexia, vomiting, diarrhea, dry mouth, sedation, lethargy, depression, decreased total thyroxine and free thyroxine in euthyroid dogs, mydriasis, changes in liver enzymes, seizures.[b,e]
Buspirone (azapirone)	0.5–2 mg/kg, PO, q 8–12 h[c,e]	Anxiety and fearfulness.[b,c,d,e] Extralabel.	Vomiting, diarrhea, nausea, agitation, behavioral disinhibition, bradycardia, stereotypical behaviors.[d,e]
Trazodone (serotonin antagonist and reuptake inhibitor)	2–7.5 mg/kg, PO, q 8–24 h as needed, up to 19.5 mg/kg in any 24-h period[e,g]	Anxiety disorders, fear aggression, compulsive disorders, sedation/anxiolysis for veterinary care, postsurgical confinement.[d,e,g,h,i] Extralabel.	Nausea, vomiting, diarrhea, sedation, ataxia, agitation, restlessness, increased anxiety, aggression, tachycardia. Hepatotoxicosis and priapism have also been reported.[d,e]
Clonidine (alpha-2-adrenergic	0.01–0.05 mg/kg, PO, q 8 h as needed, or q 8–12 h when [c,d,e]	Anxiety, fear, phobias, panic, neuropathic pain.[d,e]	Sedation, hypotension, nausea, dry mouth, ataxia, sleep disturbances, excitement,

agonist)	administered regularly[c,d,e]	Extralabel.	urinary incontinence, bradycardia, hypotension, collapse.
Dexmedetomidine oral gel (alpha-2-adrenergic agonist)	125 mcg/m^2 (not mg/kg), oral transmucosal, q 2 h as needed for maximum 5 doses during one noise event[j]	FDA approved for canine noise aversion. Extralabel indications include other fear, anxiety (eg, fear of car rides, fear of veterinary care).[d,j]	Sedation, vomiting, gastroenteritis, drowsiness, inappropriate urination.[j,k]
Gabapentin (gabapentinoid, or alpha-2-delta ligand)	5–30 mg/kg, PO, q 8 h when administered regularly, or 20–50 mg/kg, PO, 90 minutes before stressor as needed[c,d,e]	Anxiety, fear, neuropathic pain, seizures.[d,e] Extralabel.	Sedation, ataxia, vomiting, diarrhea, increased appetite, agitation, excitement.
Alprazolam (benzodiazepine)	0.02–0.1 mg/kg, PO, q 6 h as needed[b,e]	Anxiety, fear, phobias, seizures.[d,e] Extralabel.	Sedation, increased appetite, ataxia, muscle relaxation, anxiety, paradoxical excitation. Aggression has also been reported.
Diazepam (benzodiazepine)	0.5–2 mg/kg, PO, q 8 h as needed[d,e]	Anxiety, fear, phobias, seizures, appetite stimulant, muscle relaxant.[d,e] Extralabel.	Sedation, increased appetite, ataxia, muscle relaxation, anxiety, paradoxical excitation. Aggression has also been reported.

Abbreviations: SSRI, selective serotonin reuptake inhibitor; TCA, tricyclic antidepressant.

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[i] Jokela F. Reduction of repetitive tail chasing in a mixed breed dog using multimodal therapy. *J Vet Behav*. 2025;78:106-112. doi:10.1016/j.jveb.2025.03.001

[j] Sileo (dexmedetomidine oromucosal gel). Orion Pharma. Revised November 2017. Accessed September 3, 2025.

[k] Stahl SM. *Essential Psychopharmacology: The Prescriber's Guide*. Cambridge University Press; 2005.

Treatment of Normal but Undesirable Behavior in Dogs

For normal but undesirable behaviors, owners require guidance and resource material on meeting dogs' behavioral needs and understanding learning principles to reinforce what is desirable while preventing what is undesirable. Owners should be given high-quality resources about training and behavior management.

Veterinarians should counsel dog owners to avoid positive punishment–based techniques that suppress undesirable behavior but do not help a dog learn alternative, more desirable behavior choices. Techniques that cause pain or fear can lead to persistent fearfulness and anxiety, as well as to aggression. Dog trainers who use scientifically validated, force-free techniques provide important guidance to owners addressing undesirable behaviors.

When a dog's behaviors are normal but excessive, a more in-depth behavior assessment might be needed to fully evaluate the problem, formulate a

prognosis, and implement a comprehensive treatment plan. In some households, making the necessary changes to achieve a goal that meets the owner's needs might be impractical or impossible.

Treatment of Fear, Anxiety, and Aggression-Related Behavior Problems in Dogs

A treatment plan for a behavior disorders should be based on a diagnosis and prognosis that are determined in a behavior consultation. For most behavior abnormalities, resolution requires environmental management to prevent further problems, along with behavior modification.

In some cases, behavior-modifying drugs are needed to relieve the dog's distress and establish a mental and emotional state conducive to learning.

Sometimes, referral to a veterinary behaviorist (ie, a veterinarian who is a specialist in behavior and is certified by a relevant certifying board, such as the American College of Veterinary Behaviorists, European College of Animal Welfare and Behavioural Medicine, Latin American Veterinary College of Animal Welfare and Behavioral Medicine, or Australian and New Zealand College of Veterinary Scientists Veterinary Behaviour Chapter), is warranted.

Environmental Management of Fear, Anxiety, and Aggression-Related Behavior Problems in Dogs

If all of the dog's needs are adequately met, the first step in a behavioral treatment program is to identify each situation, stimulus, or interaction in which the problem might arise, so that a preventive program can be implemented. Prevention ensures safety (eg, in aggression cases), prevents further damage to the household or injury to the dog, avoids further anxiety-evoking situations for the dog, and ensures no further aggravation of the problem through fear conditioning (ie, unpleasant outcomes) and learning (ie, negative reinforcement if the stimulus retreats).

The most effective prevention is to identify and avoid any situation in which the dog might be exposed to the stimulus. Modifications in the environment (eg, distance, visual barriers, auditory barriers, physical barriers) might be required. Establishing a routine can greatly help anxiety.

When possible, it's best to introduce changes gradually.

Behavior Modification for Fear, Anxiety, and Aggression-Related Behavior Problems in Dogs

Structured interactions can be beneficial in treating problems rooted in fear and anxiety. Structured interaction programs provide the dog with consistent and predictable owner responses, which decrease anxiety and build self-control by improving communication and allowing the dog to control its consequences/outcomes.

Programs based on this idea are sometimes termed "**cue-response-reward**." Before gaining access to something it desires, the dog can choose to respond to a cued known behavior (eg, sit) and then is granted the desired treat or privilege (eg, toy, play, attention). With consistent implementation of the program, the dog learns that cued behaviors are associated with rewards.

In most cases, treatment of problems related to fear, anxiety, impulsivity, and aggression focuses on changing the dog's emotional response to the stimulus (counterconditioning) and/or replacing the undesirable response with one that is desirable by using reinforcement-based techniques (response substitution). Teaching the basics of reinforcement-based training enables owners to focus on rewarding desirable behaviors (eg, sit, look at me, relaxed down).

Clicker training can be particularly useful. Because it enables owners to mark the precise moment a desirable behavior occurs with a click, it facilitates their ability to gradually shape outcomes that more closely approximate the final goal.

Positive punishment should be avoided, because it can cause fear and anxiety toward the owner or fear of the situation and surrounding triggers.

Arousal might have to be decreased before treatment can proceed in some cases. Dogs that are highly aroused respond with autonomic fight-or-flight responses and tend to respond reflexively, interfering with their ability to learn. Arousal can be decreased by training the dog to settle on cue and by minimizing the intensity of the stimulus during exposure (desensitization).

Management devices, such as head halters, that can change the dog's focus and help the dog settle can be helpful. Early intervention with behavioral pharmacotherapy might be necessary to achieve success and can be in the best interest of a fearful, anxious, or phobic dog.

Behavioral Pharmacotherapy for Fear, Anxiety, and Aggression-Related Behavior Problems in Dogs

Psychotropic drugs and adjunctive therapies (eg, supplements, nutraceuticals, pheromones, nutritional therapies) can be used to reestablish a more stable emotional state and improve trainability in dogs that are anxious, fearful, or overly reactive. Although drugs can improve the dog's emotional state and facilitate new learning, only with concurrent behavior modification can new neuronal pathways be established, new behaviors learned, and fearful responses to stimuli changed to positive ones.

Depending on whether they are administered regularly or only situationally, medications can be classified as either maintenance medications or as-needed (rapid-onset) medications.

Baseline or **maintenance** anxiolytic medications should be considered when triggers of fear, anxiety, or aggression occur frequently and unpredictably. They might also be indicated alongside situational medications for dogs that display chronic anticipatory anxiety or that struggle to recover from exposure to stimuli that elicit fear, anxiety, or aggression.

Classes of drugs used for behavioral pharmacotherapy in dogs and cats include selective **serotonin reuptake inhibitors (SSRIs)** and **tricyclic**

antidepressants (TCAs).

Among SSRIs, **fluoxetine** is typically the first choice. In the US, a veterinary formulation of fluoxetine is approved for treatment of canine separation anxiety in conjunction with a behavior modification plan. Fluoxetine is commonly used in an extralabel fashion to treat various other behavior problems (aggression, anxiety, inappropriate urination, and compulsive behavior) in dogs. Fluvoxamine, **sertraline**, and **paroxetine** can be alternatives when fluoxetine proves insufficiently effective or when adverse effects, such as anorexia, are a concern.

Among the TCAs, **clomipramine** has the highest level of serotonergic activity. In the US, a veterinary formulation of clomipramine is approved for use as part of a comprehensive behavioral management program to treat separation anxiety in dogs > 6 months old. Clomipramine is also used as an alternative to fluoxetine for treating anxiety disorders and phobias in dogs.

Buspirone, an azapirone, can be used to enhance the anxiolytic effects of SSRIs or TCAs.

Selegiline, a monoamine oxidase inhibitor that is approved in North America for treating cognitive dysfunction syndrome in dogs, is also approved in Europe for the treatment of emotional disorders or chronic anxiety (2).

Certain medications (rapid-onset medications with a short duration of action) can be used **as needed** when a fear stimulus can be anticipated:

- **Dexmedetomidine** oral gel is an alpha-2-adrenergic agonist approved by the FDA for treating canine noise aversion. Because of its rapid onset and short duration of action, it can be particularly useful during thunderstorms.
- **Clonidine**, another alpha-2 adrenergic receptor agonist, has a slower onset and longer duration, making it helpful for prolonged exposure to frightening stimuli (3).
- **Trazodone**, a serotonin antagonist and reuptake inhibitor, is another option (4). Its onset is slower (90–120 minutes), and it has a duration of action lasting 4–6 hours.
- **Gabapentin**, which is an anticonvulsant, analgesic, and anxiolytic, is also an option for situational treatment of fear and anxiety (5). Combining gabapentin and trazodone can help further decrease anxiety and fear.
- Benzodiazepine medications such as **diazepam**, **alprazolam**, **clonazepam**, and **lorazepam** are powerful anxiolytics that can greatly decrease fear and anxiety. However, drugs in this class have been associated with paradoxical excitement and increased aggression in some instances, so they should be prescribed with caution.

Because as-needed drugs have a range of individual variability in behavioral effects and adverse effects, trials with different as-needed medications should be conducted in advance to determine which drug and dose achieve the desired effect and for how long.

Natural products can be used alone or in conjunction with drugs or other natural products to help alleviate or lessen anxiety; however, evidence of their effectiveness in dogs is somewhat limited. Several studies support the use of **pheromone therapy**.

A commercially available synthetic dog-appeasing pheromone (apasine) analogue mimics the calming pheromones produced by lactating dogs and is available as a spray, diffuser, or collar. Pheromones have been used to treat anxiety related to car rides, veterinary visits, separation anxiety, and aversion to storms and fireworks, as well as to lessen adoption stress and facilitate socialization.

A calming or anxiety-decreasing effect has been observed with alpha-casozepine (a milk protein hydrolysate), with a botanical product containing *Magnolia officinalis* and *Phellodendron amurense*, with L-theanine, and with aromatherapy (using lavender).

The essential amino acidL**- tryptophan** is a precursor to serotonin. Dietary supplementation of L-tryptophan has been investigated as a means of treating problem behaviors in dogs (6, 7, 8).

Commercial diets have been developed that incorporate L-tryptophan and alpha-casozepine or that include omega-3 fatty acids and probiotics. These diets might help to decrease stress and anxiety.

Fears, Anxiety, and Phobias in Dogs

Although the terms "fear," "anxiety," and "phobia" are often used interchangeably, a precise understanding of the differences between these terms can help clinicians form more accurate diagnoses and more effective treatment plans:

- Fear is a response to an immediate or proximate real or perceived threat. It is a normal and adaptive response in the context of and in proportion to actual danger.
- Anxiety is sustained apprehension of a threat or danger that is neither present nor near. When not excessive and not out of context, it is normal and adaptive.
- A phobia is a marked, persistent fear that is irrational and disproportional to a threat and is always maladaptive.

Contributors to excessive anxiety and fearfulness might include genetic factors and early, even prenatal, negative experiences such as illness, poor nutrition, and severe stress trauma. Insufficient opportunities to habituate to a varied environment and a lack of socialization during the socialization period of development between 3 and 12 weeks of age also increase the risk of fear, anxiety, and phobias.

Phobias of Discrete Stimuli (Animate and Inanimate) in Dogs

Dogs can develop fear or phobias of animate stimuli such as some or all unfamiliar people or unfamiliar dogs. Common inanimate triggers of fear and phobia are loud and unpredictable noises such as thunderstorms, fireworks, or gunshots; environments such as boarding kennels or veterinary clinics; surfaces such as wooden flooring; or experiences such as car rides or being left alone.

After exposure to a fear trigger, a dog can develop apprehension or anticipation that results in anxiety during situations similar to those in which the dog experienced fear. Phobic responses can result in secondary fears of associated stimuli. For example, dogs with thunderstorm phobias might become fearful of rain, wind, or changes in air pressure.

Fear responses that occur in many contexts can be characterized as **global fear**. **Generalized anxiety** is persistent anticipatory fear, difficulty settling, and hypervigilance.

Clinical signs of **acute fear** and anxiety can be behavioral and autonomic. They include tachycardia, tachypnea, excessive salivation, and GI disturbance. Behavioral clinical signs include low body posture, piloerection, vocalization, and displacement behaviors such as yawning or lip licking.

A fearful dog might choose avoidance, flight, or dispersive aggression, depending on the context, the dog's genetic predisposition, and past experience.

Treatment of Phobias of Discrete Stimuli in Dogs

Treatment of specific fears and phobias relies on managing the environment to decrease exposure to frightening stimuli, changing the dog's emotional response to the stimulus (classical conditioning), and helping the dog manage its arousal by teaching skills such as settling on cue (operant conditioning). Early intervention with medication might be necessary to achieve success and can be in the best interest of a fearful, anxious, or phobic dog.

The first step in a treatment program for fears and phobias is to identify each situation, stimulus, and interaction in which the problem might arise so that an avoidance strategy can be implemented. Avoidance has an immediate effect of decreasing stress and important therapeutic value, because each exposure solidifies fear learning, making it more difficult to overcome.

Dogs with noise phobias might benefit from soundproofing (eg, sound-absorbing curtains or panels), white noise, ear covers, or noise-canceling headphones. Developing a safe haven protected from frightening noise (eg, in a finished basement or sound-insulated crate) can be very helpful for predictable noise triggers such as thunderstorms or public fireworks displays.

For a safe haven to be effective, dog owners must build positive associations with it long before using it during actual events. When the safe haven is used for a storm or other noise event, the dog should be taken there before the frightening noises start.

Some dogs benefit from dog clothing that applies gentle compression or special metallic fabric that discharges static buildup.

The ultimate goal of behavior modification to treat phobias of discrete stimuli is to successfully expose the dog to controlled levels of the stimuli while it remains calm and free of anxiety. The dog gradually learns to tolerate a fear-triggering or arousing stimulus (eg, a frightening noise) that is introduced at a low enough intensity not to elicit a reaction, and then gradually increased to the intensity at which it typically occurs.

Counterconditioning can also be used, to form an association between the stimulus and something pleasurable, such as food. By having the stimulus paired with an enjoyable activity or object, the dog's emotional response to the stimulus can shift from fear or anger to contentment and calm.

To ensure an effective outcome, a dog owner should work with a trainer or veterinarian to design a method of counterconditoning that minimizes and controls the intensity of the stimulus.

For noise phobias, controlled exposure is best accomplished through the use of recordings. Recordings can be played at a volume low enough that a fear response is not evoked and paired with a positive stimulus for counterconditioning. Provided that the dog remains relaxed, sessions can be gradually, systematically increased in intensity (volume and duration of exposure). However, linear increases in stimulus intensity should be avoided. That is, instead of turning up the volume of the recorded noise each session, the owner should vary the volume, with an overall trend toward increasing it.

With phobias that involve multisensory experiences (eg, thunderstorm or fireworks phobia), controlled exposure to the sound (eg, of thunder or fireworks) seldom leads to full resolution of the phobia.

Flooding is a method in which an animal is made to stay in the presence of a fear-evoking but harmless stimulus until the fear is extinguished. Flooding is not recommended. Because the exposure must be continued until all physiological and emotional signs of fear are gone, it is difficult to do, and if done improperly, flooding can increase problem behaviors (sensitization). Furthermore, flooding has negative welfare implications for dogs. Because the dog is exposed to the full intensity of the frightening stimulus for a prolonged time, flooding risks subjecting the dog to severe emotional or even traumatizing distress.

Although controlled exposure behavior modification relies on classical conditioning, **operant conditioning** has an important role in promoting a successful outcome. This can involve teaching cues that enable avoidance of uncontrolled exposures to stimuli (ie, teaching the dog to move away or back up), which can decrease the risk of setbacks that occur from accidental flooding. Learning to relax on cue on a designated mat or bed helps the dog manage its arousal and fear during controlled exposure sessions, enabling more rapid progress. See the table General Guidelines for Effective Controlled Exposure Behavior Modification in Dogs.

TABLE

General Guidelines for Effective Controlled Exposure Behavior Modification in Dogs

- The dog must remain calm throughout exposure sessions.
- The dog's body language should be used as a guide to the pace of progress. The intensity of the stimulus may be increased only when the dog shows no response or a positive response to it.
- If a reaction occurs during a desensitization session, the owner should stop immediately, allow the dog to regain calmness, and then revert to a level that won't provoke a response (eg, shorten the duration) and start again.
- Frequent, shorter sessions are more effective than infrequent, longer ones.
- The owner should be counseled to avoid linear increases in intensity of the stimulus. For example, instead of turning up the volume of a recorded noise each session, the owner should vary volume both up and down, with an overall trend toward increasing it.
- If there is no progress or if the dog begins to regress, the owner should pause and reevaluate. Minor adjustments in context (such as location, time of day, type of treats, and timing) can greatly influence the outcome.

Anxiolytic drugs and supplements can be beneficial components of treatment for fear- and anxiety-based problems:

- For discrete stimuli that can be anticipated, such as fireworks, a rapid-onset medication with a relatively short duration of action, such as dexmedetomidine oromucosal gel, trazodone, clonidine, or a benzodiazepine such as alprazolam, should be considered.
- When exposure to frightening stimuli is chronic or unpredictable, a maintenance medication (TCA, SSRI, azapirone) might be needed.
- For dogs with anticipatory anxiety as well as acute fear, a maintenance medication and situational medication can be beneficial.
- Dogs with thunderstorm phobia might benefit from seasonal use of an SSRI or TCA and situational medication during storm events..

Separation Distress Disorder in Dogs

Separation distress disorder is a manifestation of physical, physiological, or behavioral fear or anxiety that arises during or in anticipation of the absence (or perceived absence) of individuals. Some dogs show distress when separated from certain specific caregivers and cannot be comforted by the presence of other individuals. Other dogs might become distressed only when they are alone and lacking human companionship.

The risk that a puppy will develop fear and anxiety when left alone can be decreased by allowing it to regularly and comfortably spend time alone during the day (preferably in a safe haven).

In some instances, separation distress disorder occurs when a dog has a frightening experience, such as hearing thunder, while left alone.

There is substantial comorbidity between noise phobia and separation distress disorder.

Clinical signs of separation distress disorder include destructive behavior (especially at exits or directed toward the owner's possessions), distress vocalization, house soiling, salivation, pacing, restlessness, inability to settle, anorexia, and repetitive or compulsive behaviors. Dogs that show signs of close attachment behaviors, such as following family members around the house, are at no greater risk of separation anxiety than dogs that do not display these behaviors (9).

Clinical signs of separation distress disorder typically occur within the first 15–30 minutes after an owner's departure. However, many dogs with separation anxiety begin to show signs as the owner prepares to leave (eg, by putting on shoes, grabbing keys, going to the door). Signs of distress have been observed to change over time, with vocalization peaking early, and pacing and orienting becoming more prominent later (10). When the owner returns, greeting responses are often exaggerated, and the dog can be difficult to calm down.

Video recording is an invaluable diagnostic tool to observe the behavior and determine whether there are other concurrent signs of anxiety (such as autonomic stimulation, increased motor activity, and heightened vigilance and scanning) or additional environmental triggers and behaviors that can indicate a different diagnosis.

A diagnosis of separation distress disorder requires the exclusion of other common causes of the signs (eg, incomplete housetraining, exploratory play and scavenging, external stimuli leading to arousal and anxiety, noise aversion, or confinement anxiety).

Treatment of Separation Distress Disorder in Dogs

Dogs with separation anxiety should be left alone as rarely as possible to prevent further sensitization, which makes treatment more difficult. For dog owners who cannot bring their dogs to work, day care or dog sitters can be beneficial.

Owners should be instructed meet the dog's age- and breed-specific environmental needs with a consistent routine for play and exercise.

Regular positive reinforcement training, which increases communication and the predictability of a dog's environment, is also an important part of treatment.

Dogs with mild separation anxiety might respond to relaxation training and counterconditioning

Pearls & Pitfalls

- A diagnosis of separation distress disorder requires the exclusion of other common causes of the signs (eg, incomplete housetraining, exploratory play and scavenging, external stimuli leading to arousal and anxiety, noise aversion, or confinement anxiety).

with food. The dog is taught to relax in a designated place where it can enjoy independent activities (chews, puzzle feeders, etc). When the owner prepares to leave, the dog is taken to the designated place and given a selection of high-value food, chews, and special toys. When the owner returns home, the special items should be removed. The aim of this simple behavior modification protocol is to build positive associations with being alone.

For dogs with moderate to severe separation anxiety, a graduated departure training protocol using safety signals is recommended. This approach increases the predictability of departures and the dog's tolerance of being alone. It differs from previous behavior recommendations geared toward decoupling predictors of departure such as picking up keys or putting on shoes.

A safety signal is a novel object or sound not associated with previous departures, which indicates that the departure will be "safe" (ie, that the owner will return before the dog becomes anxious). In daily training sessions, simulated departures of gradually increasing length are used to build the dog's capacity to remain calm and comfortable alone. In each training session, the owner places the safety signal immediately before starting the departure routine.

Simulated absences must be short enough not to trigger anxiety. In the beginning, therefore, departures should be very brief. Monitoring the dog's response to treatment with video enables the owner to make sure the dog remains under its threshold of anxiety during simulations.

Clients should be informed that the graduated departure behavior modification protocol can take months to complete (11).

Two medications—the SSRI fluoxetine and the TCA clomipramine—are approved in the US by the FDA for treatment of canine separation anxiety in conjunction with a behavior modification plan. Both have been shown to be effective at decreasing clinical signs, especially when combined with behavior modification. For dogs that do not tolerate either fluoxetine or clomipramine, other SSRIs might be good alternatives.

A rapid-onset medication should also be considered to provide situational relief for dogs that must be left alone before the daily medication takes effect and behavior modification has been completed. Options include trazodone, gabapentin, clonidine, and alprazolam, all of which take effect in 90 minutes and have a 4- to 6-hour duration of action.

Aggression in Dogs

Aggression is the use of force to cause harm (eg, injurious bites) or threat of force (eg, subtle changes in body posture, facial expression, and vocalizations). The term "aggressive" describes a dog's action in a given context and should not be used to describe a dog's temperament. The term should not be used synonymously with "vicious" or "mean".

In the animal world, aggression is a normal and adaptive behavioral response to threat that enables self-preservation. Aggression can be defensive or offensive, appropriate or inappropriate for the given context. Aggressive behavior may be considered pathological when it occurs out of scale or out of context for the situation or environment.

In referral practices across North America, aggression is the most common behavior problem in dogs, accounting for approximately 70% of the caseload (12). It is also a major human concern, because at least 5 million people are referred to the hospital each year in the US alone for treatment of dog bites.

Most forms of aggression are dispersive (ie, motivated by a desire to create distance between the dog and the target of aggression). There are many motivations for aggressive behavior in dogs, including anxiety and conflict (uncertainty).

Genetics, early development (prenatal, postnatal), socialization, and previous experience and learning all play a role in the development of aggression.

Before treating a dog for aggression, the veterinarian must evaluate for medical conditions and assess the potential risk of injury posed by the dog to humans and other animals:

- Medical conditions that might cause or contribute to aggression must be identified, because they are important factors in diagnosis, prognosis, and treatment.
- All stimuli that might incite aggression should be accurately identified to ensure initial safety. Predictability is a critical issue in prognosis, both to prevent further incidents and to develop a stimulus gradient for treatment.
- The signalment, environment, history, and target of the aggression provide valuable information about whether the problem might be safely and effectively managed.
- The type of aggression should be identified. Some types can be managed and improved; others require prevention.
- The dog owner's understanding and ability to effectively and safely prevent the problem and comply with the treatment recommendations must be assessed.

Risk factors and prognostic indicators should be discussed with the owner. Safety is a serious concern, and the prognosis is generally guarded to poor for dogs showing aggression that is unpredictable, disproportional to the level of threat, uninhibited, resulting in substantial injury, or directed toward individuals that cannot reliably avoid the dog (eg, children). The European Society of Veterinary Clinical Ethology provides risk assessment guidelines.

To minimize immediate risk of injury and ensure that treatment can be effective, the veterinarian and dog owner must first manage the dog's anxiety and arousal by avoiding all known and suspected triggers of aggression and contexts in which aggression is more likely to occur (eg, interactions that increase the dog's arousal). Dogs with elevated arousal are at high risk of aggression because their decision-making is affected by their physiological

state (ie, flight or fight).

Reward-based training should be initiated or restarted. Positive reinforcement improves communication between the owner and the dog and provides a foundation for behavior modification that will facilitate avoidance of triggers and change the dog's response to inciting stimuli.

Owners should be cautioned to avoid confrontation and punishment, which can heighten the dog's arousal, anxiety, and fear, thus increasing the risk of aggression.

Medications can help achieve a behavioral state conducive to new learning. In some cases, notable improvement might be achieved with a combination of careful environmental management, behavior modification, and medication.

Because aggression can lead to liability and poses a public health risk, general practitioners should refer to or consult with a board-certified veterinary behaviorist in cases of moderate or severe aggression.

Fear-Related Aggression in Dogs

Fear is the most common motivation for aggression in dogs. Fear-related aggression is aggressive behavior to ward off a perceived threat. Aggression is used to increase the distance between the dog and the frightening stimulus.

Genetic factors, early (even prenatal) negative environmental influences (such as illness, excessive stress, poor nutrition, or inadequate socialization), or trauma at any age can lead to fearfulness and fear aggression. Learning also plays an important role in the development and maintenance of fear-related aggression. Dogs can learn from experience that escape or avoidance is not possible, whereas aggression can effectively remove a threat.

Over time, fear-related aggressive behavior might appear offensive, as the dog learns to display aggression proactively in anticipation of a threat. For example, owing to a history of attempts to avoid contact being thwarted, a fearful dog might begin to growl whenever a stranger tries to pet it. With continued exposure, the dog might begin growling when it sees strangers at a distance.

A diagnosis of fear-related aggression is based on identifying signs of fear, as well as noting a history and progression of fear and aggression. Physical health problems that cause discomfort or pain should be ruled out, because they can lower a dog's threshold for fear and aggression.

Treatment of Fear-Related Aggression in Dogs

Avoiding stimuli that trigger aggression is critical for safety and for preventing further reinforcement of fear and aggressive behaviors. For example, dogs that display aggression toward strangers should be confined behind a gate or on leash when unfamiliar people visit the owner's home. Dogs that exhibit aggression on leash should be walked in areas where they are less likely to encounter dogs or people that frighten them. A basket muzzle can be useful in situations where triggers of aggression cannot be entirely avoided.

Behavior modification should include desensitization and counterconditioning to stimuli that elicit aggression.

SSRIs and TCAs can increase the success of behavior modification in dogs by decreasing fear and impulsivity. Trazodone or clonidine can also be used situationally for stressors such as the arrival of guests, either in lieu of or in addition to daily medication.

Redirected Aggression in Dogs

Dogs direct aggression toward a third party when they are prevented from exhibiting or unable to exhibit aggression toward their primary target. In the most common form of this aggression, the dog bites an owner who is grasping or restraining it when trying to prevent or break up a dog fight. Similarly, dogs that might be aggressive toward a veterinarian might bite the person restraining the dog.

Redirected aggression arises as a result of the frustration or interruption of other forms of aggression or arousal.

Treatment of Redirected Aggression in Dogs

Treatment of redirected aggression requires identifying and treating the underlying cause of fear or arousal.

Territorial Aggression in Dogs

A dog might display aggression when approached by people or other dogs within its territory. Territory can be stationary (eg, yard, home) or mobile (eg, car). A dog's perception of its territory might extend beyond the boundaries of its owner's property, to include, for example, neighbors' yards.

Whereas territorial aggression is defined as aggression that occurs only on a dog's perceived territory, aggression in this context more commonly results from contextual fear. The space (eg, a small entryway) might make it difficult for the dog to choose escape when a visitor enters. And the dog's aggressive response might be due to learning (eg, the dog learns that only aggression works to prevent unwanted petting that typically occurs in that location).

Fear, anxiety, defensive, and possessive behaviors can also contribute to territorial aggression in dogs. When a dog is on its own ground, its motivation to escape or avoid (flight) might be decreased or absent.

Aggressive behavior can sometimes appear to be territorial, because the stimuli eliciting aggression are more likely to occur at home. For example, a

fearful dog might pass strangers calmly on walks but bark and growl when a stranger comes to the home because there's a history of the dog's receiving unwanted attention from visitors and of being ignored by strangers on walks.

Learning (negative reinforcement when the stimulus retreats) and fear conditioning (unpleasant outcomes such as yelling, discipline, electric shock, etc) can also play a role.

Treatment of Territorial Aggression in Dogs

To prevent territorial aggression, owners should be counseled to prevent their dog from having access to people and dogs entering their property. The dog should be confined when visitors come to the home. Doors and gates should be kept locked to avoid unexpected encounters with visitors. The yard should be surrounded by a physical fence.

Tie-outs and electric fences should not be used, because they do not hinder people from entering the dog's territory. One study found that dogs within an electronic pet containment system might be at increased risk of aggression toward people on or near the property because of avoidance-motivated aggression (13).

Behavior modification should focus on minimizing the dog's arousal in the presence of visitors and on teaching a behavior such as remaining quiet on a mat when visitors come to the house.

SSRIs or TCAs can be helpful for dogs with comorbid fear, anxiety, hypervigilance, or excessive arousal. Trazodone or clonidine might also be useful on an as-needed basis for fear, anxiety, and arousal associated with time-limited events such as the arrival of a visitor.

Conflict Aggression in Dogs

Conflict aggression occurs when a dog is anxious, uncertain, or conflicted during an interaction and uses aggression to control the outcome of the confusing or undesirable interaction. Many dogs that display conflict aggression are easily aroused, impulsive, anxious, or excessively fearful. Aggression can be elicited by confronting the dog, handling it in an unwanted way, interrupting a desirable activity, or attempting to remove a valued resource.

When dogs are resting or sleeping, chewing on a favored object, or no longer desirous of human affection, they might respond with either deferential displays or threats. However, if the owner continues to approach, tries to remove the resource, or attempts to pet the dog despite its signaling, aggression can escalate, and future signaling can be lost.

Inconsistent or unpredictable human-dog interactions can cause emotional conflict in the dog, in which the dog might desire a relationship but does not trust that the interaction will be safe.

Because children are less likely to understand canine social signaling used to express discomfort, they are more likely to elicit conflict-related aggressive behavior.

Administering treatments that a dog might find aversive, such as topical ear medication, is a common trigger for conflict-related aggression. It is therefore incumbent on the veterinarian to ask the owner whether administering the treatment at home can be done safely and with undue stress for the dog. If not, help with low-stress handling and cooperative care can be offered. In some cases, an alternative treatment might need to be considered.

Genetic factors, early-life negative experience, and insufficient socialization as puppies contribute to the development of conflict aggression in dogs.

Learning also plays an important role in conflict aggression. When a dog successfully uses aggression to achieve a goal (eg, retaining a resource) or to remove a threat, it learns that aggression is successful (negative reinforcement).

Because conflict aggression can be due to many factors (eg, prior learning, fear conditioning, underlying conflict), it can be difficult to determine the dog's underlying motivation for the conflict.

Aggression directed toward family members has been mislabeled as dominance or status-related aggression. The term "dominance" describes a relationship between two individuals of the same species within a social group, as reflected by actions, interactions, and intraspecific communication/signaling. These relationships are not established by aggression of the dominant individual but rather by the deferential signaling of others.

One study found no evidence of a dominance hierarchy in dogs and thus between dogs and human companions (14). The investigators concluded that what appeared to be dominant/deferential behaviors between individuals were described more accurately and parsimoniously by associative learning and consideration of the value of resources over which conflict occurred.

Aggression toward family members generally arises from fearful or defensive behaviors, resource guarding, redirected behavior, or situations of conflict (competing emotional states and unpredictable consequences) or to an owner's attempts to inhibit excessive play aggression. Owners who respond to conflict aggression with physical techniques that are intended to assert dominance (eg, pinning the dog or rolling it over) and verbal discipline (yelling "no") can sometimes cause the dog to inhibit further aggression; more often, though, such techniques cause increased fear and anxiety, and thus increased aggressive behavior.

Treatment of Conflict Aggression in Dogs

Environmental management to support avoidance of stimuli that cause conflict aggression should be instituted immediately for safety and to avoid reinforcing aggressive behaviors:

- Items that could be stolen should be kept out of reach.
- Food, high-value chews, long-lasting treats, and toys that trigger aggression should be provided in an out-of-the-way location where the dog will not be disturbed (eg, in a crate or pen).
- Beds and crates should be placed in areas where the dog can rest undisturbed.
- All unwanted handling should be avoided.
- Owners should avoid confrontation, because it can contribute to a loss of trust, increase in anxiety, and escalation of aggression.

Positive reinforcement training enhances communication and increases the predictability of interactions between owner and dog, which decreases anxiety and conflict. Behavior modification can be used to increase the dog's comfort with people approaching while it has an object of value. Owners can toss the dog high-value treats whenever they approach or pass by the food bowl and offer high-value treats or toys whenever the dog voluntarily gives up a toy or chew. In addition, the dog can be trained to give and drop items on cue (beginning with items of low value for high-value rewards).

For dogs that are sensitive to touch, a consent petting protocol should be used. Handling for care is best addressed with cooperative care training, which is a positive reinforcement–based approach in which dogs learn to cooperate with nail trimming, grooming, ear cleaning, and other required care in exchange for food rewards.

SSRIs can decrease the anxiety and impulsivity that contribute to conflict-related aggressive behavior and facilitate behavior modification.

Possessive Aggression (Resource Guarding) in Dogs

Dogs exhibit possessive aggression when approached by a person or another animal while they are in possession of something they wish to retain or when they are near a valued object. Aggression is most commonly displayed when the dog is near to or in possession of highly motivating food, treats, chew toys, stolen items, resting places, or even preferred individuals.

Some behaviorists categorize aggression aimed at retaining possession of a valued resource as conflict aggression; others categorize it separately as possessive aggression.

Genetics and early experience play a role in the development of possessive aggression. However, context and learning are important. The relative value of the object to the dog and the perceived risk associated with guarding it from another dog or person also affect whether aggression might occur. The dog might learn that it can successfully retain the object with aggression.

Fear and defensive behavior also play a role, if the owner threatens, punishes, or confronts a dog that resists handling or guards an item.

Treatment of Possessive Aggression in Dogs

The treatment for possessive aggression is the same as for conflict aggression.

Aggression Toward Other Dogs

Aggression toward unfamiliar dogs and those that are not members of the family group is likely motivated by fear, resource guarding, frustration, or territoriality. Agonistic behavior between dogs living in the same household can be caused by resource guarding, fear and anxiety, arousal, impulse dyscontrol, overarousal, or poor intraspecific communication skills.

In some cases, aggression toward a canine housemate might be redirected fear or frustration. Anxiety, fearfulness, impulsivity, and poor canine communication skills can be the result of genetics, lack of early socialization, or negative experiences and can be compounded by learning.

Competition among dogs over valued resources is usually resolved without aggression, via an exchange of social signals. Aggression can occur if owners inadvertently support or encourage a dog to hold on to a resource during an encounter in which the dog would normally defer. Age or illness can also play a role, if the way in which one dog signals or responds to another is altered.

The risk of interdog aggression within a household is highest in female-female pairs and lowest in male-female pairs (15, 16). Research on the effects of neutering on interdog aggression is inconclusive (17).

Treatment of Aggression Toward Other Dogs

Environmental management aimed at avoidance is essential for safety and for the successful treatment of aggression between familiar dogs.

Resources that trigger aggressive interactions should be made unavailable when the dogs are together. Dogs should be separated during meals and when given high-value chews, toys, etc. The dogs might also need to be separated when arousal is likely to be high (eg, when visitors arrive or when the owner comes home).

Interactions such as rough play that can trigger fear or anxiety might also have to be avoided.

Owners should be taught to recognize canine social signals so that they can intervene and separate the dogs before aggression escalates. In severe

cases, the dogs might need to be leashed or wear basket muzzles when together, or they might need to be separated entirely.

Owners should be instructed to implement reward-based training to improve communication, as well as predictability and consistency of interactions. Implementing a cue-response-reward protocol in which dogs are asked to perform a simple behavior (such as to sit on cue) before desirable activities and treats is also helpful, because it encourages the dog to check in with the owner (16, 18).

Dogs in the same household should be taught to go to separate designated places on cue. A place cue facilitates avoidance and enables owners to quickly separate the dogs, if they observe social signals that indicate an increased risk of aggression (18).

Treatment with an SSRI can be a useful adjunct to behavior modification. In one study, fluoxetine was associated with positive outcomes (16). Other SSRIs (eg, sertraline, paroxetine, citalopram) and clomipramine can also be effective at treating aggression motived by fear and anxiety (19, 20, 21).

Play Aggression (Inappropriate Play Behavior) in Dogs

In dogs, play is a normal, ritualized behavior that can persist into adulthood as a result of genetics and learning. Although play postures and interactions might mimic agonistic behavior, they lack an intent to threaten or harm, which distinguishes this behavior from aggression.

Pouncing, wrestling, nipping, and biting exhibited during play rarely escalate to aggression; however, these behaviors might become excessive and lead to injury. If a dog becomes overly aroused during play, there is an increased risk of aggression if something occurs to change the dog's emotional state.

Treatment of Play Aggression in Dogs

To prevent the development or persistence of undesirable play behaviors with people, clear and consistent rules should be communicated about acceptable play behavior. Teaching puppies to sit before receiving anything of value (eg, toys, treats, play) encourages them to look to the owner for direction. Puppies should be engaged in regular acceptable forms of play, including fetch, tug games, and playing with manipulation and chew toys.

When undesirable play behaviors (eg, oral behaviors, or "mouthiness") occur, interaction should be immediately stopped (ie, negative punishment). A verbal cue can be used to interrupt overly rough or mouthy play. If needed, a leash and head halter can help to interrupt undesirable play. To avoid causing frustration, negative punishment should be immediately followed by redirection to an alternative, more desirable activity (eg, redirecting the puppy to a more acceptable form of play).

Positive punishment (eg, scolding or pushing away) should not be used to stop play, because it can lead to fear of the owner, defensive aggression, or conflict-induced aggression. For some puppies, positive punishment inadvertently reinforces undesirable play behavior.

Predatory Aggression in Dogs

Predation is a normal canine behavior but can be dangerous. Predatory aggression is one of the most dangerous types of aggression in dogs, because there is usually no warning. The attack is intended to kill prey, and the bite is uninhibited. The sequence of events might include stalking, chasing, biting, and killing.

Young children, babies, and small animals can be at risk of predatory aggression by dogs, because their size and behaviors mimic those of prey. Although extensive socialization to a species might decrease predation toward that species, the behavior can be enhanced when predatory individuals are together in a group.

Treatment of Predatory Aggression in Dogs

Dogs that exhibit excessive predatory behavior or predatory behavior directed toward inappropriate targets must be managed carefully to avoid all exposure to triggers of the predatory behavior. Sometimes, when a dog's predatory behavior is directed toward cats, the dog cannot safely live together with cats.

Safety is a serious concern. Before treating a dog for aggression, the veterinarian must assess the potential risk of injury that the dog poses to humans and other animals.

The prognosis is generally guarded to poor for dogs showing predatory aggression, which, by its nature, is uninhibited and likely to result in substantial injury and is often directed toward individuals (eg, children) that cannot reliably avoid the dog. In cases of predatory aggression toward people, especially children, referral to a board-certified veterinary behaviorist is recommended; euthanasia might need to be considered.

Maternal Aggression in Dogs

Maternal aggression in dogs can occur in intact females with a litter of puppies or in females with pseudocyesis. It can be directed toward people or toward other animals. Signs of aggression arise when the bitch is protecting puppies or toys that mimic puppies, and the aggression should resolve when the hormonal state returns to normal and/or the puppies are weaned.

The term "maternal aggression" has also been used to describe aggression or cannibalism directed toward puppies by the bitch. Although this problem might have a genetic component, it is reported to occur more frequently after a first litter.

Treatment of Maternal Aggression in Dogs

Ovariohysterectomy can prevent further incidents of maternal aggression.

Pain-Induced/Irritable and Medically Caused Aggression in Dogs

Any disease that causes pain or discomfort—pruritus, nausea, etc—can increase irritability and aggression in dogs (1, 22, 23). Aggression can be a direct response to pain or discomfort. A painful dog might display defensive aggression when it is handled or anticipates being handled (1).

Pain and discomfort can also exacerbate other behavior problems associated with aggression, including fear-related aggression, conflict aggression, possessive aggression, or territorial aggression (1).

Organ (eg, renal, hepatic) dysfunction, CNS disease, and endocrinopathies (eg, hyperadrenocorticism, functional testicular and ovarian tumors, and thyroid dysfunction) might also contribute to irritability and aggression.

It has been suggested that hypothyroid dogs might display an increase in aggression; however, the data are inconclusive at this time (22, 24, 25).

Treatment of Pain-Induced/Irritable or Medically Caused Aggression in Dogs

Successful resolution of pain-induced/irritable or medically caused aggression requires diagnosis and treatment of the underlying physical health problem. Although necessary, the resolution of physical health problems might not be sufficient for full resolution of aggression. Fear learning that occurred while the dog was experiencing pain might result in subsequent defensive aggression when the dog is handled or approached after a painful physical health problem is resolved.

Clinicians should not delay treatment of fear, anxiety, and distress until investigation of the physical health problem has been completed (22). Environmental changes at home—avoiding handling or rough play, moving the pet's bed away from a children's play area, etc—might provide the dog some immediate relief from fear, anxiety, and stress associated with illness and pain.

Behavior modification (desensitization and counterconditioning) should be recommended to treat persistent aggression caused by fear learning.

Abnormal Repetitive Behaviors in Dogs

Abnormal repetitive behaviors occur on a spectrum that includes stress displacement behaviors, stereotypies, and canine compulsive disorder. Neurological, dermatological, GI, and other medical pathologies can also be present with abnormal repetitive behavior disorders.

Stress displacement behaviors are normal behaviors that occur out of context and serve to restore a dog to emotional homeostasis during times of heightened stress or frustration. Common displacement behaviors include tail chasing, spinning, pacing, scratching, and digging. They are usually self-limiting and tend to resolve quickly when the cause of stress or frustration is gone. Frequent occurrence of stress displacement behaviors might indicate persistent anxiety or a problem in the environment.

When an environment persistently fails to provide sufficient outlets for engagement in normal behavior, stereotypies can develop. Stereotypies are repetitive behaviors that tend to be unvaried in sequence. Like other abnormal repetitive behaviors, stereotypies have no obvious purpose or function, but they can, at least in the early stages, provide a coping mechanism for the dog. Stereotypies might be induced by dopaminergic stimulation.

Canine compulsive disorder might first arise as a displacement behavior when the dog is frustrated, conflicted, or highly aroused. Lack of predictability in the daily routine, alterations in the environment, unpredictable consequences, lack of sufficient outlets for normal behaviors, and chronic or recurrent anxiety might be initiating factors.

The diagnosis is considered to be a compulsive disorder when the behavior occurs to an extent that it interferes with normal function or when it becomes independent of (or emancipated from) the inciting stimulus. With canine compulsive disorder, there is likely altered serotonin transmission.

Compulsive disorders are repetitive, stereotypical, locomotory, grooming, ingestive, or hallucinogenic behaviors that occur out of context to the time and situation in which they take place and occur at an excessive frequency or duration. There might be lack of control over onset or termination. Although it can be debated whether animals can obsess, dogs do perceive and experience concern; therefore, the term "obsessive-compulsive" has also been used to describe such disorders.

Because there is likely a genetic component for many compulsive disorders, the signalment and age of onset are important. For example, German Shepherd Dogs and Bull Terriers are at higher risk of spinning or tail chasing, and Labrador Retrievers are more prone to excessive licking of distal forelimbs. A genetic locus for flank sucking has been identified in Doberman Pinschers (26).

As with other behavior problems, diagnosis of a compulsive disorder is a diagnosis of exclusion. It is therefore necessary to rule out other problems before confirming a diagnosis of canine compulsive behavior.

The diagnostic process should start with a careful history and observation of the behavior. Video recordings might be needed if the behavior is not exhibited during the consultation. Because many physical health problems can cause compulsive-like clinical signs, a thorough workup should be done to rule out physical health problems before a diagnosis of canine compulsive disorder is made.

Differential diagnoses depend on the behavior exhibited:

- For ingestive behaviors such as pica, licking, sucking, swallowing, and jaw snapping (sometimes called "fly biting" or "fly catching"), differential diagnoses associated with GI or metabolic diseases, including gastric ulcers, inflammatory bowel disease, food insensitivity, hyperadrenocorticism, and hypoadrenocorticism, should be considered.
- For compulsive disorders associated with licking, sucking or self-mutilation—such as flank sucking, acral lick dermatitis, tail mutilation, psychogenic alopecia, or nail biting—atopic dermatitis, bacterial or parasitic hypersensitivity, adverse food reaction, parasitic infestation, infection, neuropathy, autoimmune disease, and pain from any cause should be investigated.
- Tail chasing might indicate sacral or caudal orthopedic or neurological pain.
- Spinning, star gazing, pouncing, or light or shadow chasing might indicate a neurological problem such as focal seizures or other neuropathy.

A clinicopathological database (CBC, serum biochemical analysis, and urinalysis) should be obtained to identify possible renal, hepatic, and metabolic abnormalities. Endoscopy, ultrasonography, a food trial, a trial of a gastric protectant, or a steroid trial might be required to rule out GI problems.

To investigate causes of pain or discomfort that could trigger licking or mutilating of a body part, radiography, ultrasonography, or MRI might be needed. For suspected dermatological problems, skin scraping, trichography, food trial, allergy testing, fungal culture, or biopsy should be considered. Empirical treatment for atopic dermatitis (eg, with oclacitinib or lokivetmab) may be considered.

An MRI might also be useful for ruling out certain neurological diseases.

Treatment of Abnormal Repetitive Behaviors in Dogs

The first step in treating abnormal repetitive behaviors in a dog is to make sure that the dog has age- and breed-appropriate exercise and mental stimulation.

Environmental management can decrease the dog's exposure to triggers of compulsive behaviors. For dogs that chase light and shadows, caregivers can close the shades during the time of day when the contrasts between light and shadow are greatest.

Behavior modification can be used to provide the dog with an **alternative behavior that is incompatible with the repetitive behavior** during times when the repetitive behavior is most likely to occur. For example, the dog can be taught to relax in bed, where it will be given a long-lasting treat or chew.

Owners can be counseled to gently interrupt compulsive behavior and **redirect the dog to alternative activities**, such as playing or chewing. If needed, a head halter can be used to guide the dog's head away (eg, from the site of a lick granuloma) before directing the dog to an alternative activity.

Because most dogs respond to drugs that inhibit serotonin reuptake, including **SSRIs and clomipramine** (a serotonergic TCA), treatment with SSRIs or clomipramine should be a first line of pharmacotherapeutic treatment. However, alterations of other neurotransmitters (eg, dopamine, endorphins, N-methyl-D-aspartic acid [NMDA]) can play a role, so augmentation with an NMDA blocker such as memantine might be helpful if SSRI or TCA therapy is not adequately effective.

When physical examination, history, and diagnostic testing do not clearly identify the cause of the behavior, a therapeutic response trial might be indicated (eg, anticonvulsants to treat suspected focal seizures as a cause of jaw snapping or light chasing). If anxiety contributes to the repetitive behavior, augmentation with trazodone might also be helpful (27).

Behavior modification, environmental management, and behavior-modifying drugs might decrease the frequency and intensity of repetitive behavior in dogs with canine compulsive disorder. A full resolution of clinical signs, however, is rarely possible.

Hyperkinesis (Hyperactivity) in Dogs

Hyperkinesis in dogs, which is analogous to human attention-deficit/hyperactivity disorder (ADHD) is associated with impulsivity, excessive motor activity, and an altered ability to acquire new tasks (learn), sometimes accompanied by stereotypical behaviors or aggression (28). There can also be signs of sympathetic activity (eg, increased heart and respiratory rates, vasodilation) even at rest.

Hyperkinesis is thought to be a result of a gene-environment interaction (28, 29).

Diagnosis can be based on history and clinical observations. Validated questionnaires can also be helpful (28, 30). However, consensus on a diagnosis of hyperkinesis in dogs is currently lacking in the veterinary community (28).

Differential diagnoses include normal behaviors in dogs with behavioral needs that are not being adequately met, or compulsive, anxiety, or impulse control disorders.

Treatment of Hyperkinesis in Dogs

Dogs with hyperkinesis should be provided adequate opportunity for breed- and age-appropriate exercise and environmental enrichment. A consistent routine, which increases predictability and can lower a dog's arousal, should be implemented. Situations in which the dog is likely to become overstimulated should be avoided.

Owners should use positive reinforcement training, offering frequent and timely rewards to keep the dog interested and focused. Positive punishment

and negative reinforcement should be avoided, because they can lead to fear and anxiety.

Relaxation protocols such as relaxed down stays can be helpful (28).

Destructive Behaviors in Dogs

Many of the destructive behaviors of dogs, including chewing, stealing, garbage raiding, and digging, are normal exploratory behaviors that become destructive when the dog is unsupervised and not otherwise engaged in more desirable activities.

Some dogs engage in destructive behaviors because of fear or anxiety (eg, due to confinement or noise phobias). When destructiveness occurs only in the owner's absence, separation anxiety should also be considered. Together with the history, videotaping or camera monitoring is generally the best way to diagnose behavior problems that occur when the owner is absent, as well as to assess the dog's response to treatment.

Treatment of Destructive Behaviors in Dogs

Environmental management is often sufficient treatment of destructive behaviors in dogs. A regular daily routine with sufficient reward training, exercise, and social enrichment can help to ensure that these behaviors do not arise when owners are present; however, varying degrees of supervision might be required.

When owners cannot supervise, the dog should be provided with adequate outlets for exploratory play in the form of chews, food-stuffed toys, or manipulation toys. Outdoor dogs might also benefit from a designated area for digging.

Dogs that engage in destructive behaviors might also need to be confined away from areas where problems could arise, or they might need to be housed in crates, pens, or runs to prevent access to potential targets of destruction.

Eating Disorders

Canine behavior problems related to ingestion include those in which food intake is excessive (polyphagia), inadequate (hyporexia), or too fast (gorging); water intake is excessive (polydipsia); or nonfood items (pica) or feces (coprophagia) are eaten.

Medical causes should be excluded first. GI abnormalities can trigger pica, hyporexia, and food aversions. Metabolic diseases (eg, Cushing disease, diabetes insipidus) can cause polyphagia or polydipsia. Medications such as corticosteroids and phenobarbital can also increase appetite and lead to polyphagia.

Many dogs, especially puppies, begin to chew and ingest nonfood items as part of investigative and exploratory behavior. Some dogs that scavenge do so as a normal component of food acquisition and are reinforced by success. Dogs with hyporexia might have an anxiety disorder, and some might develop specific taste preferences and aversions that decrease what they will eat. Some dogs with pica and polyphagia have compulsive disorders.

Dogs with a history of excessive confinement and insufficient stimulation might learn to ingest or eat feces. This coprophagia might be a normal behavior in some cases. Maternal behavior includes consumption of feces and urine of young puppies. In addition, as part of exploratory behavior, many dogs are attracted to and might ingest feces, compost, and prey (dead or live). Coprophagia occasionally has a medical cause.

Treatment of Eating Disorders in Dogs

Many feeding problems in dogs can be improved by a program in which the dog is given food as a reinforcer for training (sometimes called a "work-for-food" program), with the balance placed inside toys that require chewing or manipulation to release the food. This strategy encourages exploration; makes feeding an enjoyable, time-consuming, and mentally challenging activity; and can limit the quantity of food consumed and prevent gorging.

As with most behavior problems, correction of feeding problems in dogs needs to be accompanied by management strategies to prevent access to potential targets whenever the dog cannot be supervised or actively engaged in other chew and play behaviors.

Treatment with SSRIs or clomipramine might helpful for compulsive disorders involving consumption.

Elimination Behavior Problems

House Soiling in Dogs

Dogs might eliminate (urinate or defecate) in inappropriate locations because of inadequate or insufficient training, as a marking behavior, or as a result of fear or anxiety. However, pain, sensory decline, cerebrocortical disease (including cognitive dysfunction), or any medical condition that leads to increased volume of urine, more frequent elimination, pain on elimination, or lack of control, must first be excluded as potential causes or contributing factors.

House Soiling in Dogs Due to Inadequate or Insufficient Training

House soiling in dogs can be due to inadequate or insufficient training. It can also be a result of fear, anxiety, or arousal.

House soiling can also be caused by pain, impaired mobility, sensory decline, cerebrocortical disease (including cognitive dysfunction), or any medical condition that leads to increased volume of urine, more frequent elimination, pain on elimination, or lack of control. Therefore, physical health problems must be excluded as potential causes or contributing factors.

Barriers to outdoor elimination (frightening outdoor stimuli such as rain, fireworks, neighbor animals, etc) should also be considered as causes of or contributors to house soiling.

A detailed behavior history is necessary to determine whether the dog has been fully housetrained. Dogs obtained from locations where they were extensively caged (eg, pet stores) and dogs that have lived exclusively outside might be more difficult to housetrain, because they have never had to inhibit elimination.

Treatment of House Soiling in Dogs Due to Inadequate or Insufficient Training

If housetraining was absent or insufficient in a case of house soiling by a dog, a housetraining regimen should be discussed. The owner must accompany the dog to a designated outdoor elimination area, reinforce elimination immediately with a food reward, and supervise the dog indoors to prevent opportunities for elimination (eg, by keeping the dog on leash).

The dog should be taken outside at regular intervals. Elimination is most likely to occur after meals, upon waking from sleep, and after finishing an activity such as playing. The dog should also be monitored for signs that it is ready to eliminate, such as sniffing around, standing near the door, or attempting to sneak out of view of the owner, and should be taken out immediately if these signs are observed.

When the owner is not able to supervise, a combination of scheduling (ensuring that the dog eliminates before the owner's departure, having someone come in to take the dog outside, etc) and confinement is required. Dogs can either be confined away from areas where they might eliminate or kept in an area where they will not eliminate, such as a pen, room, or crate where they eat, play, or sleep. Alternatively, the dog can be provided with an indoor elimination area (eg, paper, indoor puppy potty) within a confinement area where it can relieve itself when the owner is gone.

Behavior modification should focus solely on reinforcement of elimination in desirable locations. Punishing a dog after finding evidence of house soiling is ineffective because the dog will not associate the punishment with having eliminated in the house previously, and the punishment might thus cause anxiety.

Punishing a dog caught in the act of eliminating in the house is also counterproductive. It can trigger a fear of eliminating in the presence of the owner, which will impede housetraining. If the dog is observed while eliminating inappropriately, the owner should use a neutral interrupter (eg, clapping hands or whistling) and then take the dog outside. When the dog resumes urinating or defecating outside, it should be rewarded.

Marking Behavior in Dogs

Although marking occurs most often in intact males, as a form of social and olfactory communication, it also occurs in females (especially when in heat) and in neutered males and females, often to overmark other odors.

Common targets of urine marking are places that other pets have urinated, as well as items (such as blankets) that have the residual odor of other dogs, people, or cats. Some dogs will mark when they visit unfamiliar households, especially when another dog's odor is present.

A typical posture of a raised or partially raised leg is common when the surface to be marked is vertical. Fecal marking is uncommon.

Although marking is unacceptable when it occurs indoors, it is likely a component of normal communication.

Treatment of Marking Behavior in Dogs

Neutering intact males decreases marking behavior, and good supervision can prevent or inhibit most marking. As with inadequate training, dogs should be confined away from areas that might be marked when owners are not able to supervise.

Marking related to anxiety might be decreased by identification and treatment of the cause, perhaps with the aid of drugs or nutraceuticals that decrease anxiety.

SSRIs and TCAs might be effective at decreasing the frequency of marking (31).

Elimination in Dogs Due to Excitement, Submission, or Conflict

Dogs might release urine or feces when frightened or startled, or when overly excited, such as when greeting people. Some dogs urinate when showing deferential postures (eg, crouching to the ground or turning over to expose the belly) or when highly aroused.

Because loss of urine control can be associated with a concurrent desire to both greet and show deferential behavior, many cases are due to conflicting behavioral motivations.

Treatment of Elimination in Dogs Due to Excitement, Submission, or Conflict

Treatment of elimination that is due to excitement, submission, or conflict in dogs should focus on avoiding the stimuli (reaching, approach, eye contact) that incite the behavior and avoiding any positive punishment during greeting, which would add to fear and conflict behaviors.

Acceptable alternative behaviors that are incompatible with excitable greeting or deferential postures can be taught. Examples include a relaxed sit or any game or “trick” the dog might have learned, such as giving a paw or fetching.

By increasing sphincter control, phenylpropanolamine might be helpful in cases of excitement-induced urination (32).

Other Elimination Disorders in Dogs

Dogs with separation anxiety or other fears and phobias (eg, thunderstorm phobia, firework phobia) might eliminate when exposed to fear and anxiety-inducing stimuli or situations. Cognitive dysfunction might also lead to house soiling.

Treatment should focus on the underlying behavior problem.

Aging and Cognitive Dysfunction in dogs

Normal Aging Change in Dogs

Sensory, emotional, and cognitive changes occur in dogs as a normal part of aging.

Myopia and cochlear degeneration can result in hearing and eyesight deficits (33). Old dogs might also experience atrophy of the olfactory epithelium, which can result in a mild decrease in scent discrimination (34).

Cognitive changes include slower learning, decreased retention, and a decline in nonsocial attention (35, 36).

Emotional changes have also been noted. Aging dogs cope less efficiently with distress and display increased dependence on human family members (37, 38).

Cognitive Dysfunction Syndrome in Dogs

In addition to normal aging changes, pathological changes in cognitive function (memory, learning, perception, awareness, emotions) can occur in dogs. These changes can manifest as one or more of a group of clinical signs. Such changes are sometimes referred to by the acronym DISHAA—for disorientation, interactions, sleep-wake cycles, house soiling, anxiety, and activity changes (which can be either decreased or increased and repetitive). In addition, anxiety, agitation, and altered responses to stimuli are frequently reported in cases of **cognitive dysfunction syndrome (CDS)**.

The first and most prominent sign of brain aging is a decline in learning or memory, which is generally impractical for pet owners to assess. Neuropsychological testing of older dogs has documented memory decline beginning at 6–8 years of age and learning deficits by 9 years of age (36).

Prevalence of Cognitive Dysfunction Syndrome in Dogs

In one study, 28% of dogs 11–12 years old and 68% of dogs 15–16 years old showed signs consistent with CDS (39). An internet survey estimated CDS in 14.2% of dogs > 10 years old, with prevalence increasing with age; however, > 85% of these CDS cases had not been diagnosed (40).

Owners should be informed of the importance of reporting any signs they observe. Veterinarians should take a proactive approach in asking owners about behavior at each visit. Early detection of CDS signs provides the best opportunity to bring about improvement and slow the decline of cognitive function.

CDS in dogs is analogous to the early stages of Alzheimer disease in humans, both in clinical signs and in brain pathological changes. Some changes to brain tissue include accumulation of beta-amyloid plaques and vulnerability to oxidative damage.

As with humans, some dogs show minimal to no clinical impairment with age, whereas others develop varying degrees of deficits.

Treatment of Cognitive Dysfunction Syndrome in Dogs

ENVIRONMENTAL MANAGEMENT OF COGNITIVE DYSFUNCTION SYNDROME

Treatment of CDS in dogs should focus first on **environmental enrichment** (both physical and mental stimulation), which has been shown to slow cognitive decline and improve the signs of CDS.

Modifying the dog’s home environment helps to accommodate the limitations of old age and cognitive decline. Providing a consistent and predictable routine decreases the stress caused by change and provides needed social contact. Removing clutter and using barriers (eg, gates, leashes, fences) help to restrict access to particular areas of the home (eg, stairs) or yard to prevent injuries or escape.

Mobility can be improved by providing ramps or nonskid flooring on slippery steps. Dogs with difficulty going through a pet door can be taken outside though an ordinary door. Tactile, sound, and odor cues can help with orientation.

Dogs that can no longer be relied upon to signal when they need to go out to eliminate might need to be taken out more frequently and restricted to easily cleaned areas when the owner is not available.

The dog's day-night cycle can be reestablished by interacting with the dog during the day, providing a consistent place and time for sleeping, increasing

the dog's exposure to light during the day, and darkening the room at night.

Environmental enrichment can help delay cognitive decline by protecting against hippocampal neuron loss, increasing neuronal plasticity, and building up a cognitive reserve.

Regular exercise should be provided in accordance with the dog's comfort and mobility.

BEHAVIOR MODIFICATION FOR COGNITIVE DYSFUNCTION SYNDROME

Although more patience might be needed with dogs that have CDS, reinforcement-based training enables owners to focus on rewarding desirable behaviors and provides mental stimulation and social connection.

Response substitution (training an alternative desirable behavior) can be a useful approach to decreasing problem behaviors like pacing or barking. Reinforcement of alternative behaviors can include rewarding lying still instead of pacing, rewarding staying quiet instead of barking, and rewarding eliminating outside instead of inside.

Positive punishment should be avoided because it causes anxiety, which exacerbates the dog's decreased ability to learn.

BEHAVIORAL PHARMACOTHERAPY FOR COGNITIVE DYSFUNCTION SYNDROME

Medications, diets, and supplements might improve signs of CDS or slow its progression in dogs.

The monoamine oxidase B inhibitor **selegiline** can improve CDS signs by enhancing dopamine and other catecholamines in the cortex and hippocampus and by decreasing free radical load.

Propentofylline, which is licensed in Europe and Australia for the treatment of dullness, lethargy, and depressed demeanor in old dogs, can increase blood flow and inhibit platelet aggregation and thrombus formation.

A number of diets, nutraceuticals, and supplements have also been shown to improve CDS signs and potentially slow cognitive decline.

Commercially available diets include one supplemented with fatty acids, antioxidants, and DL-alpha-lipoic acid and L-carnitine to enhance mitochondrial function, and a specialized diet that uses botanical oils containing medium-chain triglycerides to provide ketone bodies as an alternative source of energy for aging neurons.

Other supplements and nutraceuticals that have demonstrated efficacy in improving cognitive function include a product containing a combination of phosphatidylserine, resveratrol, and vitamins E and B_6; a product containing phosphatidylserine in combination with alpha-lipoic acid, carnitine, fatty acids, glutathione, and other antioxidants; *S*-adenosylmethionine; and apoaequorin, a calcium-buffering protein found in jellyfish.

Key Points

- **It is important for veterinarians to distinguish normal but undesirable behaviors from abnormal behaviors that are due to underlying behavior disorder.**
- **When diagnosing behavior problems, veterinarians must rule out physical health problems, especially pain.**
- **Veterinarians should provide advice on management to avoid exposure to triggers of fear, anxiety, or aggression, in order to minimize the risk of injury and to avoid aggravating fear and anxiety with further fear conditioning.**
- **Psychotropic drugs and adjunctive therapies (supplements, nutraceuticals, pheromones, nutritional therapy) can be a helpful part of a treatment plan for behavior problems.**

For More Information

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- Also see pet owner content regarding behavior problems in dogs.

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